

CBCT

Three-dimensional dental X-rays

Cone beam computed tomography shows teeth and jawbone in three dimensions. We use it when additional information can answer a specific question about your diagnosis or treatment.

What a CBCT scan shows

CBCT creates a three-dimensional dataset using X-rays. We can examine cross-sectional images in different planes. In German, this examination is called **dentale Volumentomographie (DVT)**.

A small dental X-ray or panoramic image projects three-dimensional structures onto one plane. CBCT can often separate their spatial relationships more clearly, for example when structures overlap in a conventional image.

When an additional scan may help

Impacted or displaced teeth

Clarifying the position relative to neighbouring roots or the nerve canal, assessing possible root damage and planning access for exposure.

Implants and bone grafting

Assessing bone and neighbouring structures in unclear or complex situations; obtaining data for planned guided implant placement.

Root canal treatment and apicoectomy

In selected cases, examining complex root anatomy, the extent of inflammation or proximity to neighbouring structures.

Other findings within the bone

For example, assessing the extent of a cyst or another jaw lesion when this may influence treatment.

The clinical question determines the scan. CBCT is not an automatic routine before every operation, implant or orthodontic treatment. We review existing images and explain what additional information is needed.

Benefits and limitations

CBCT can support diagnosis and planning. It does not reliably prevent all complications and does not replace examination or clinical judgement.

Metal, including fillings, implants or brace components, can create image artefacts. Movement can also reduce diagnostic quality. Soft tissues and nerves themselves are only shown to a limited extent; the bony boundaries of a nerve canal may be visible, for example.

Radiation and careful selection

CBCT uses **ionising X-rays**. A small additional long-term radiation risk cannot be entirely excluded. Before every scan, the expected health benefit must outweigh the radiation risk.

The dose depends on the equipment, size of the scanned area and imaging protocol. CBCT cannot generally be equated with the low dose of a single dental X-ray. Comparisons with medical CT also require matching equipment and protocols.

We select the smallest area sufficient to answer the question and a suitable imaging protocol. Good preparation and keeping your head still help prevent avoidable repeat scans. Any repeat examination requires a fresh assessment of its necessity.

Children adolescents and pregnancy

Children and adolescents: the need for radiation is assessed particularly carefully, and the examination is adapted to the clinical question and body size. CBCT is not an automatic part of routine orthodontic diagnostics.

Please tell us before the scan if you are or might be pregnant. We assess whether it can be postponed or replaced. If medically urgent, it may still be performed following an individual benefit–risk assessment.

Available alternatives

Depending on the question, examination, existing images, a small dental X-ray or a panoramic image may be sufficient. Other methods may be more appropriate for certain soft-tissue questions. We choose the method that answers the question with reasonable effort and risk.

If you do not wish to have a recommended CBCT scan, we discuss alternatives and any uncertainty that remains. Sometimes the planned treatment must be changed or postponed.

Clarifying costs beforehand

In German dental care, CBCT is generally **charged privately as an out-of-pocket service** for patients with statutory health insurance. We explain the specific fee and included services before the scan.

Private insurers, civil-service reimbursement schemes and supplementary insurers decide coverage according to the contract and medical justification. We cannot guarantee full reimbursement. A referral does not automatically mean that the scan is covered.

What happens during the appointment

1. **Preparation.** Bring your referral and existing X-rays or arrange for them to be transferred beforehand. A planning stent is needed only if we specifically request one.
2. **Positioning.** Depending on the equipment, you sit or stand. We position your head carefully. Remove detachable metal objects around your head and neck, glasses or dentures as instructed. Fixed braces and implants stay in place.
3. **Scanning.** The equipment moves around your head. Keep still during the short scan and follow our instructions. Tell us beforehand if positioning or keeping still is difficult.
4. **Reporting and discussion.** Expert assessment of the whole scanned volume takes longer than the scan itself. We explain the findings and what they mean for further treatment.

For CBCT alone, you usually do not need anaesthesia, an injection or contrast medium. Fasting is not necessary. Any operation or sedation at the same appointment has its own instructions.

Frequently asked questions

What should I do after the scan

After CBCT alone, you can eat, drink and continue normal daily activities. No radiation remains in your body. Collection from the practice is usually unnecessary for the scan alone.

What happens if unexpected changes are found

Changes outside the original area of interest may also be found. Such incidental findings are not automatically dangerous, but may require further assessment or specialist advice. We agree how the report and digital images will be made available for ongoing care.

Appointment checklist

- Your referral and existing images are available.
- The clinical question and costs have been discussed.
- A possible pregnancy has been disclosed.
- Any difficulty keeping still or positioning has been discussed.
- You know who will discuss the findings and continue your treatment.

Questions about CBCT: MKG im Carree · 06151 786540 · info@mkg-im-carree.de.

Problems after an additional operation outside consultation hours: +49 151 10437450; if unavailable, dental emergency service 01805 60 70 11. For a life-threatening emergency, breathing difficulty, severe swallowing difficulty or rapidly increasing swelling of the floor of the mouth, neck or tongue, call 112 immediately.

Clinical sources: [AWMF 083-005, Dental CBCT, version 3.0, December 2022, valid until December 2027](#); [German Dental Association, dental radiology recommendations](#); [Section 120 of the German Radiation Protection Ordinance](#); [Guy's and St Thomas': Dental cone beam CT scan](#). Accessed 9 September 2026. This information supplements your personal consultation.